Chapter 7: Perturbation Theory

Quantum Mechanics

1 Perturbation Equation

Definition 1.1 (Unperturbed State). [def:unperturbed-state] For a Hamiltonian $\hat{H}$ where we have a known solution to the time independent Schrodinger equation, $\hat{H}|\psi_n\rangle = E_n|\psi_n\rangle$, the *unperturbed state* is the state $|\psi_n\rangle$ with energy E_n. Such $\hat{H}$ is called the *unperturbed Hamiltonian*.

Definition 1.2 (Perturb Expansion). [def:perturb-expansion] For a Hamiltonian $\hat{H} = \hat{H}_0 + \lambda\hat{H}'$,

2 First-Order Perturbation Theory

3 Second-Order Perturbation Theory

4 Degenerate Perturbation Theory